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Description - Contains theory for the head coil. Sheet numbers listed at the end of some subsection titles refer to the sheet number within the TPS/RF (1.0T) block diagrams.

1- INTRODUCTION (HEAD COIL)

Radio frequency (RF) coils used in Signa MRI systems either transmit RF energy into the nuclei (of atoms) of a patient, receive the small MRI signal from the patient, or both. This received signal (raw data) is later constructed into an image. There are different RF coils, depending on the application being performed.

Some RF coils can be used to both transmit and receive, as with the body, head, and extremity coils. Surface coils are designed to receive only. Basic RF coil design concepts include quality factor, loading effects, and decoupling.

1-1 Quality Factor (Head Coil)

The quality factor (Q) is typically a measure of the bandwidth of an RF coil. The Q is calculated by dividing the center frequency (F_c) of the RF coil response, by the frequency difference (ΔF =bandwidth) between the half power points (-3dB from peak power) on a response curve. The RF head coils are designed for an unloaded (empty) Q of approximately 200. This means that the ΔF is approximately 320 kHz (with a center of 64 MHz). See Illustration L1937A . The higher the Q, the narrower the bandwidth. The higher the Q, the more selective the coil is. This selectivity is actually a band-pass filter that reduces the amount of noise and spurious signals that are received by the RF coil and passed on to the receiver hardware.

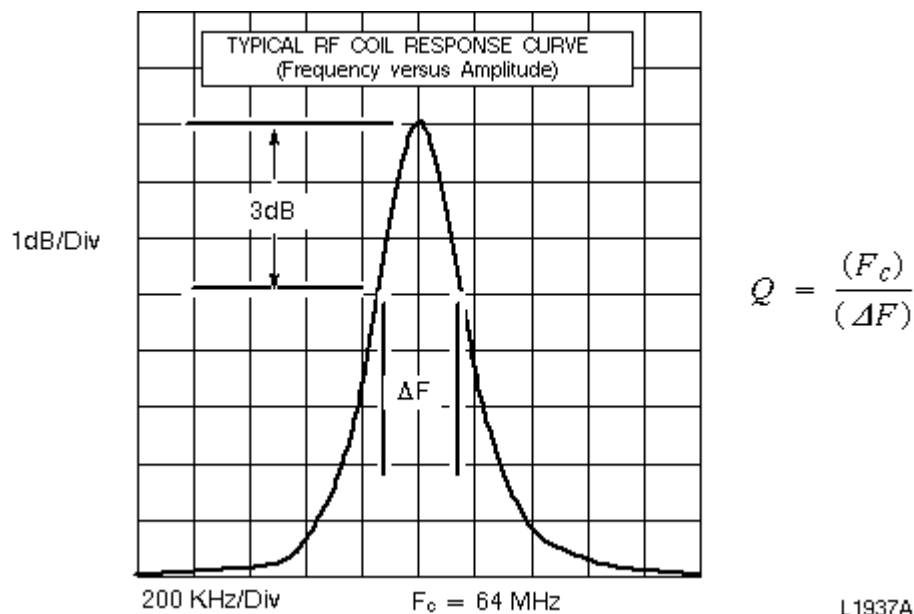


ILLUSTRATION L1937A
RF COIL QUALITY FACTOR

1-2 Loading Effects (Head Coil)

The Q is affected by the load on the RF coil. A load can be an image quality phantom, body, head, or an extremity. Different loads with different compositions (water versus muscle versus fat) have different loading characteristics. Larger loads of the same composition have more of an effect than smaller ones. The placement of a load in respect to the RF coil also has a great effect on the Q. When an RF coil becomes loaded, its response broadens, and the Q decreases, as shown in Illustration L1938A. The decreased Q results in less selectivity by the coil which, in turn, reduces the level of the received signal.

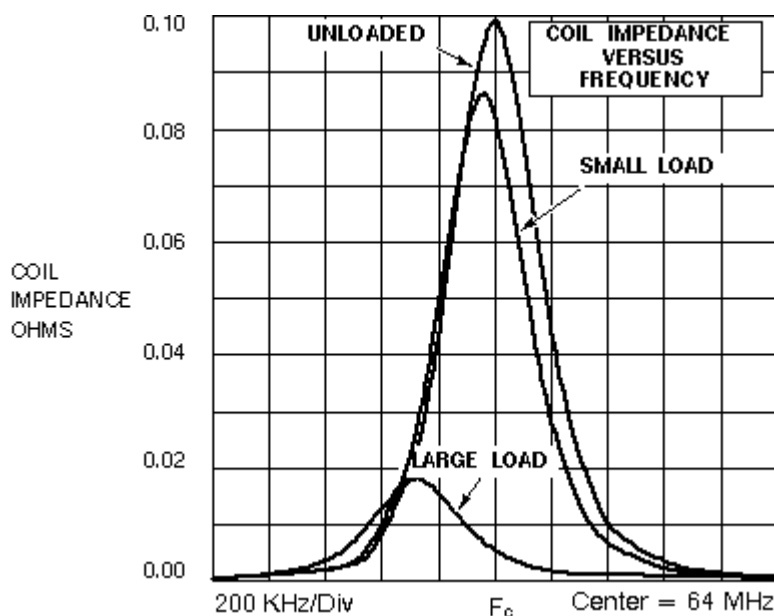


ILLUSTRATION L1938A
RF COIL LOADING EFFECTS

A frequency shift is also associated with the loaded response. With large loads, however, because the Q has decreased (and the bandwidth has increased by approximately six times), the RF coil is still, for all practical purposes, operating at an acceptable frequency. If the RF coil were tuned for a loaded condition, then a smaller load, which does not shift the RF coil frequency as much, and does not broaden the bandwidth, would not be operating at an acceptable frequency. This would cause poor image quality. RF coils are designed to be tuned unloaded. This produces the best compromise for all types of loads.

The load on a coil is very sensitive to the position of the patient with respect to the RF coil. The body acts as a capacitor. This capacitance can add to that of the coil, and can shift the frequency down. In a quadrature coil, this capacitance can affect one mode (vertical or horizontal) of the coil to the point where it is no longer operating in quadrature (a rotating RF field). It is then operating in a linear mode, which requires twice as much power to prescan, and reduces the signal-to-noise ratio (SNR) by 40%. This effect is minimized by small pads placed between the patient and the coil. The pad is actually a small capacitor in series with the body's large capacitance; this effectively cancels the body's effect.

If an RF coil is used only to transmit, while the receive function is performed by another coil, there must be a mechanism to decouple the transmit coil from the receive coil during the receive time. Without decoupling, the receive area is increased, and the benefits of a having a smaller, localized coil are compromised via noise increases and SNR decreases.

1-3 Receive Coil Design (Head Coil)

The goal of a receive coil is to maximize the SNR. Noise is a function of temperature, resistance, and system bandwidth. Because resistance increases with an increase in the coil receive area, an ideal coil is one that is only as large as is needed to scan a specific part of the anatomy. This is why receive-only surface coils have a better SNR. They are smaller and are customized for specific anatomy (e.g., neck, eye, and breast surface coils).

Quadrature design is also important for a receive coil. The quadrature effect increases the signal level, and decreases the noise level. This is due to the signal being correlated (in phase) while the noise is uncorrelated (random, or not in phase) in the rotating RF field. This improves the SNR for a quadrature coil over a linear one by a factor of 1.4.

Receive-only coils should be designed so that they are detuned during the transmit signal. This is accomplished via a *blocking network*. When the blocking network is working, the RF coil frequency response is altered so that it no longer resonates at the transmit frequency. If the blocking network is not functioning, several things can happen, including: flip angles can not be properly calibrated, the surface coil preamplifier can be damaged due to high signal levels, and the patient can be exposed to high concentrations of RF from the surface coil.

2- RF COIL DESIGN (HEAD COIL)

The discussion of the basic ladder design and RF fields can be applied to linear and quadrature coils, whether they be for body or head. All MRI RF coils are based on these two building blocks. Specific hardware pertaining to the tuning and operation of the RF coils is in the TPS/RF tabs. Refer to them for additional information.

2-1 Ladder Networks (Head Coil)

The body and head Coils used for the Signa system are based on the *ladder network* design. A ladder network can be classified as either high-pass or low-pass. A ladder network that has series capacitors and parallel (shunt) inductors is considered high-pass (see Illustration L1939A). Conversely, a ladder that has series inductors and parallel (shunt) capacitors is low-pass. A real ladder network usually has both capacitive and inductive elements in all sections, but one of the elements always dominates.

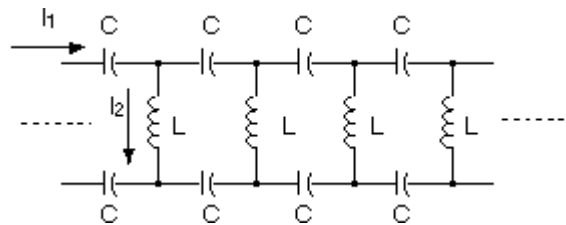


ILLUSTRATION L1939A
NORMAL HIGH-PASS LADDER NETWORK

If a high-pass ladder network is folded onto itself, the resultant elements are twice the capacitance and one half the inductance. A ground plane also occurs from the new configuration (see Illustration L1940A).

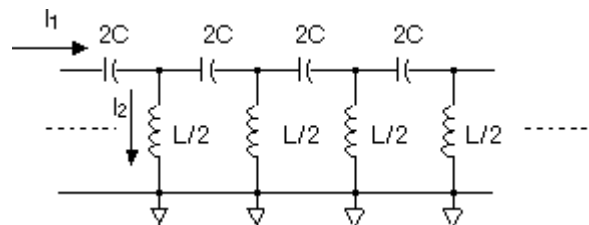
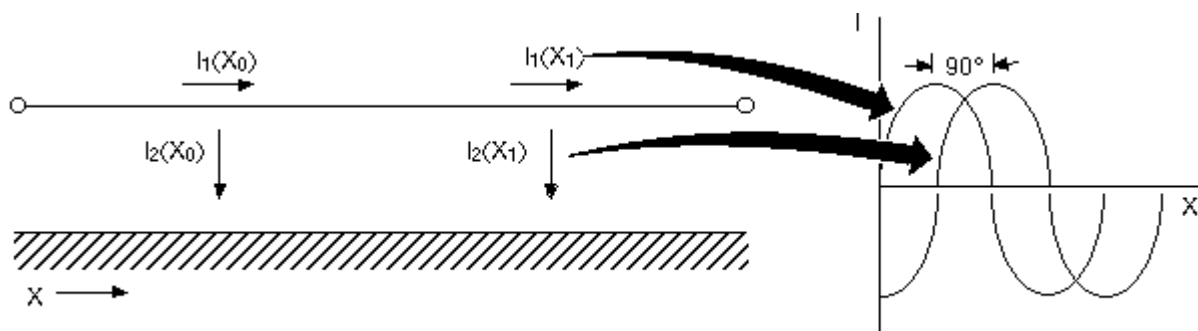


ILLUSTRATION L1940A
FOLDED HIGH-PASS LADDER NETWORK

This new ladder has the characteristics of a lumped element transmission line. The series and shunt currents in this transmission line vary along the length. The function of this varying current is sinusoidal along the length (X) of the transmission line. The series and shunt currents are also out of phase with each other. Series currents that vary as $\sin(x)$ have shunt currents that vary as $\cos(x)$, or 90 degrees out of phase with each other (see Illustration L1941A).



L1941A

ILLUSTRATION L1941A
TRANSMISSION LINE EQUIVALENT

2-2 Bird Cage Coil (Head Coil)

The quad head coil can be thought of as a high-pass ladder network that is wrapped around a cylindrical form with its ends connected together. This design looks much like a bird cage (see Illustration L1942A). The series capacitors form a ring at the ends of the coil and the parallel (shunt) inductors are horizontal runs. The actual physical elements in a quad head coil are chip capacitors placed at the end rings and copper runs on a substrate material.

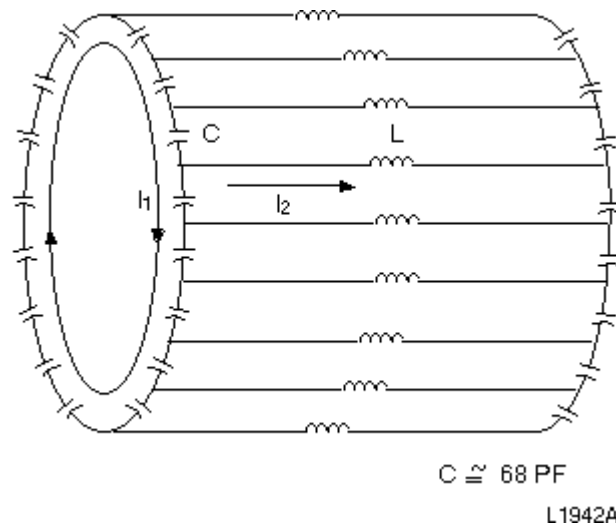


ILLUSTRATION L1942A
HIGH-PASS QUAD COIL DESIGN

Because a quad RF coil (body or head) is a ladder network with both ends connected, certain conditions must exist. One is that the current in one element must equal that in the next. Therefore, a sinusoidal (or RF) wave must be continuous. This sets up an environment where only certain frequencies of a sinusoidal wave can exist (or be resonant). In the head coil, this resonant frequency(s) is a function of both the chip capacitors placed at the end rings, and the length (or inductance) of the bird cage. The fundamental resonant frequency in the head coil is $\sin(x)$, with others at $\sin(2x)$, $\sin(3x)$ and so on. The spectrum of a typical quad head coil looks like that in Illustration L1943A .

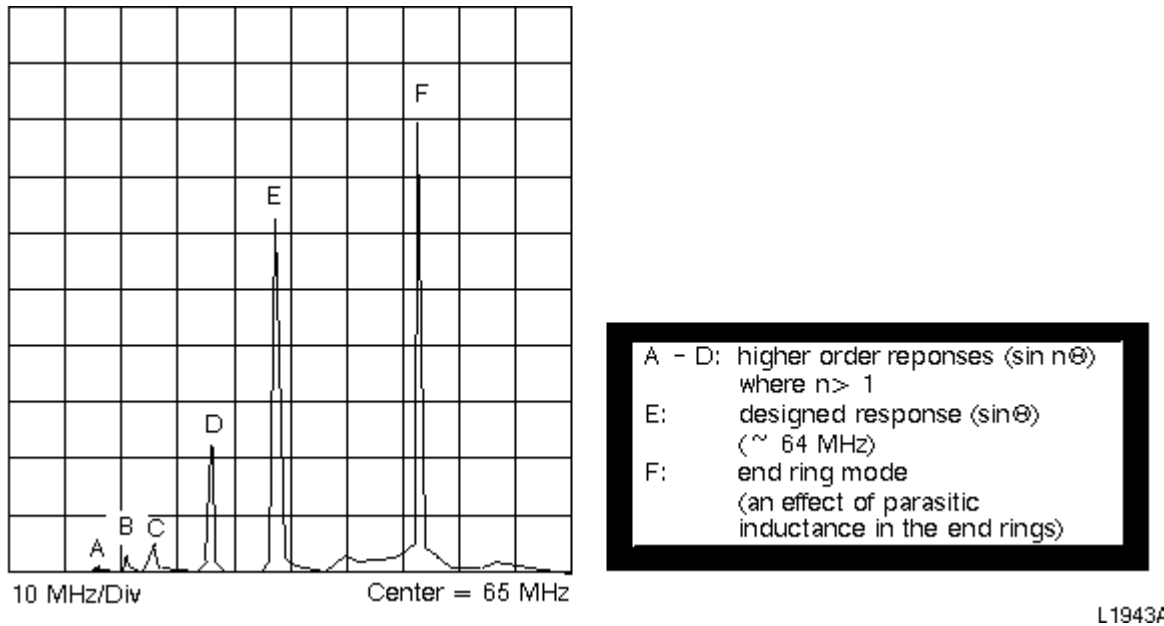


ILLUSTRATION L1943A
HEAD COIL FREQUENCY RESPONSE

2-3 Quadrature Operation (Head Coil)

A quadrature-designed coil uses two RF inputs, with one delayed by 90 degrees (or 1/4 wavelength); both are connected to the coil 90 degrees from each other, which produces a rotating RF field (see Illustration L1944A).

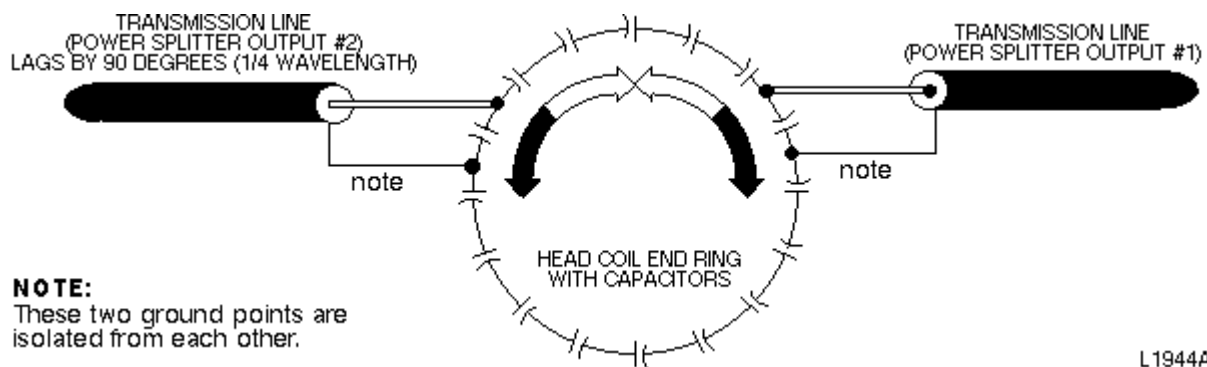


ILLUSTRATION L1944A
QUADRATURE HEAD COIL RF INPUT SIGNALS

This design is based on the fact that destructive fields (ones that do not help the MR process) cancel each other, leaving only the x and y components that produce a rotating RF field inside the coil (see Illustration L1945A).

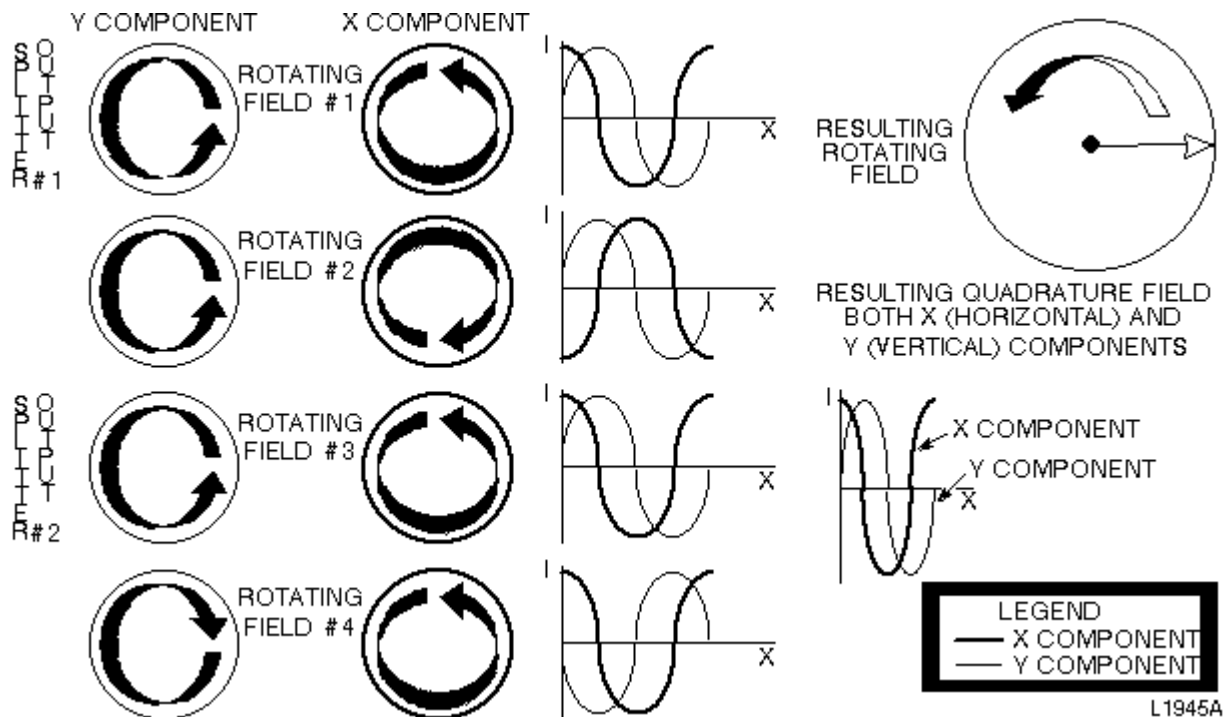


ILLUSTRATION L1945A
QUADRATURE COIL RF FIELDS

3- QUAD HEAD COIL - SHEET 9A, 9B

The quad head coil consists of three distinct parts: the hybrid splitter, two isolation boards, and the actual RF coil assembly.

This hybrid splitter is essentially the same as the body version. It splits the RF signal into two signals that are ninety degrees out of phase with each other. These two signals (I and Q) are applied to the inputs of the head coil, where a rotating RF field is produced.

The two isolation boards match the transmission line to the RF coil.

4- QUAD HEAD COIL T/R SWITCH - SHEET 9A, 9B

The head T/R is functionally the same as the Quadrature Drive Body Coil T/R switches. The T/R switches are PIN diodes that are forward biased during the transmit cycle. This allows the transmitted RF signal to reach the head coil, and short the input to the head coil preamplifier.

During receive, the same PIN diodes are reverse biased, which disconnects the RF amplifier from the head coil. The head preamp is also connected to the head coil. This ensures that maximum signal is available to the receiver.

5- HEAD PREAMPLIFIER - SHEET 9A, 9B

The echo from the patient is extremely small in amplitude. To prevent it from being obscured by noise, its level must be boosted before it is transmitted along the coaxial cable from the exam room to the receiver. The preamplifier does this. Identical units are provided for each coil in the exam room. The head coil preamplifier is also used for surface coils. For convenience, the power for each preamplifier is provided via its output coaxial cable from the inputs of the TPS receiver module. Since only one input at a time is selected in the receiver, only one preamplifier receives power. Thus, operating power and preamplifier selection are provided with no additional penetrations into the exam room. The gain of each preamplifier is ~33 dB.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Aug 27, 1998	R. Hawthorne	Initial conversion to Word
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.